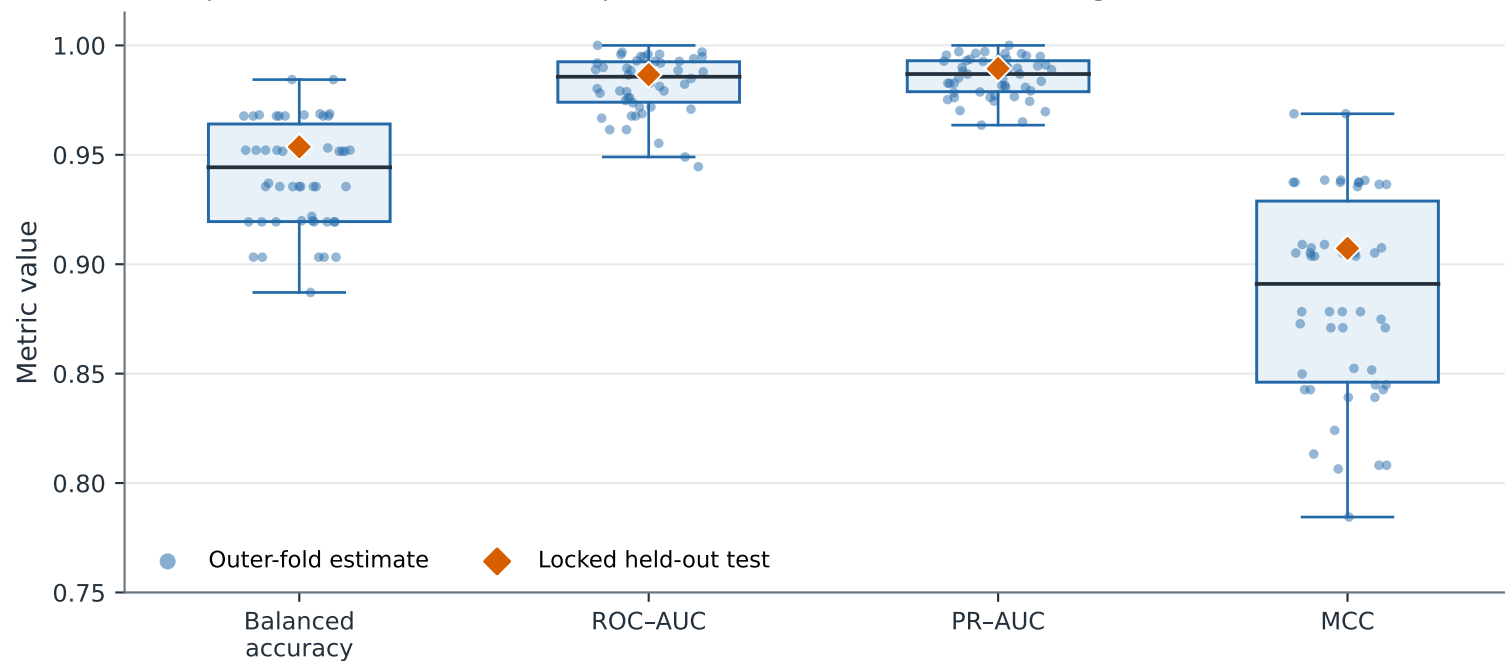


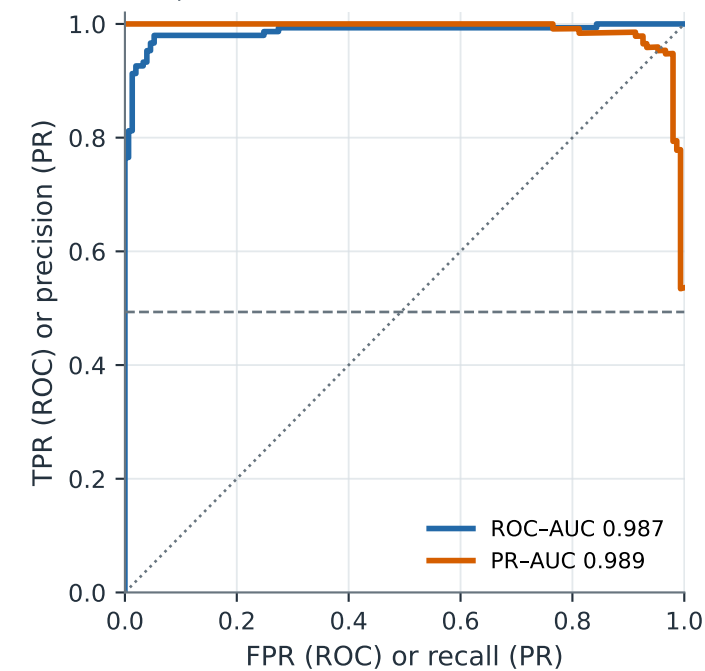
A Repeated nested cross-validation

Development set n=312; 5 folds × 10 repeats; model selection confined to training folds



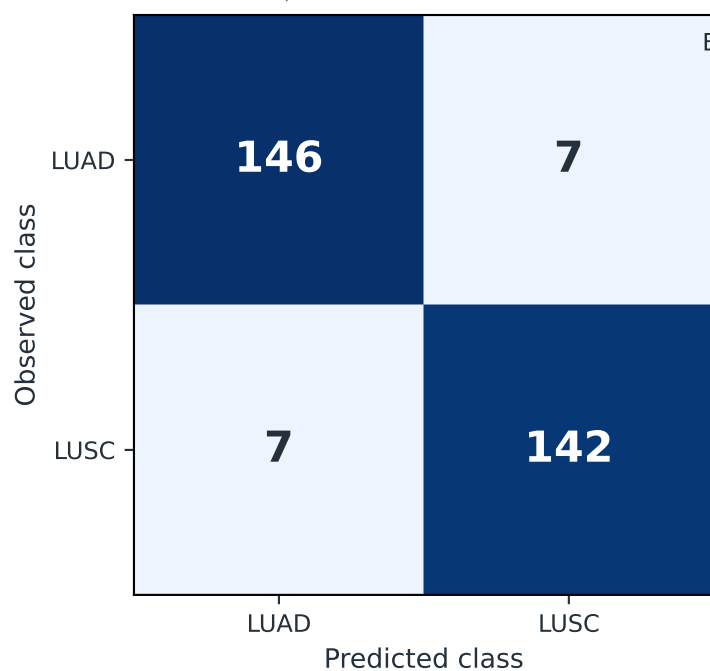
B Held-out discrimination

Independent test n=302



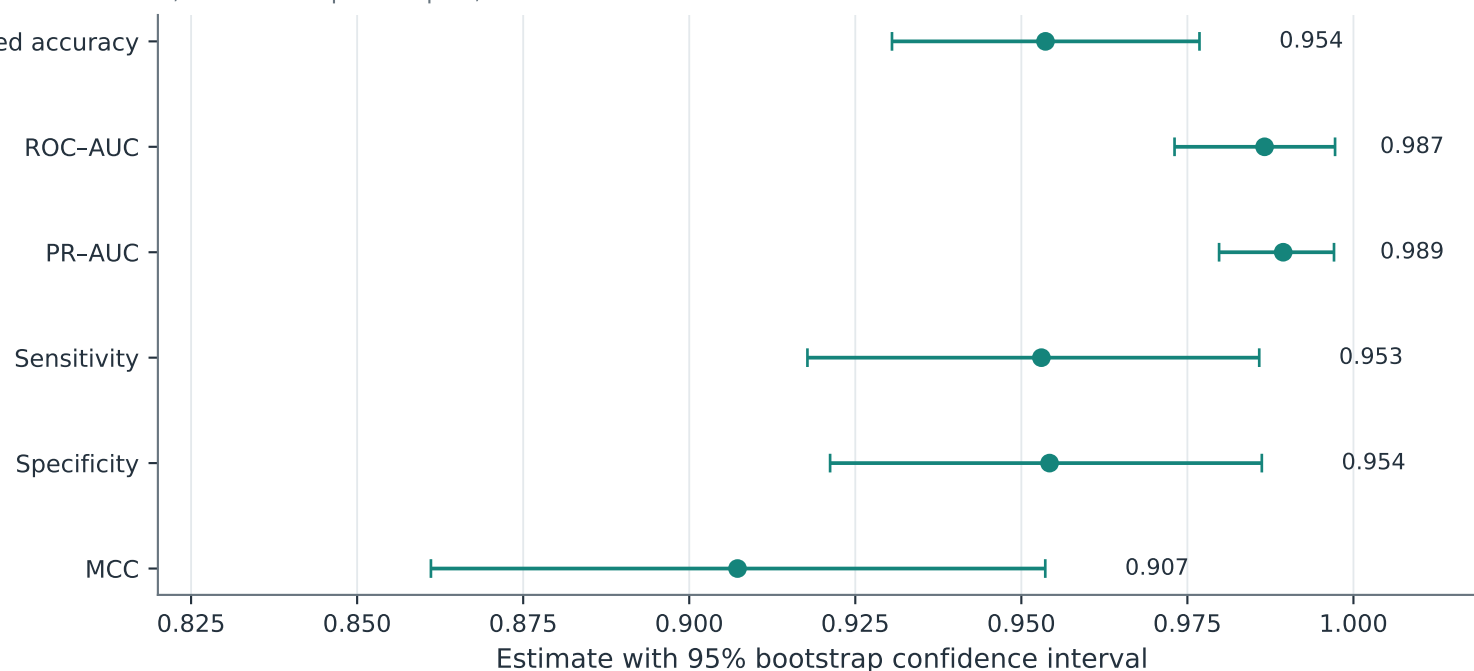
C Held-out confusion matrix

LUAD n=153; LUSC n=149



D Held-out performance uncertainty

1,000 bootstrap resamples; estimates were not used for model selection



Model: class-weighted logistic regression; fold-local median imputation, z-score scaling, and univariate feature selection. Positive class for ROC/PR: LUSC.